Arjuna JEE 2025

Maths

Trigonometric Equations

DPP: 5

Q1 If $0 \leq x \leq 2\pi$, then the number of solutions of the equation $\sin^8 x + \cos^6 x = 1$ is

(A) 2 (B) 3

(C) 4 (D) 5

Q2 The solution of the equation $(\sin x + \cos x)^{1+\sin 2x} = 2, -\pi \leq x \leq \pi$, is

(A) $\frac{\pi}{2}$

(B) π

(C) $\frac{\pi}{4}$

(D) None of these

Q3 The number of real solutions of the equation $(\sin x - x)\left(\cos x - x^2\right) = 0$ is

(A) 1 (B) 2

(C) 3 (D) 4

Q4 Equation $\sin^6 \mathrm{x} + \cos^6 \mathrm{x} = \mathrm{a}^2$ has real solution, if

(A) $\mathrm{a} \in (-1, 1)$

(B) $\mathrm{a} \in \left(-1, \frac{-1}{2}\right)$

(C) $\mathrm{a} \in \left(\frac{-1}{2}, \frac{1}{2}\right)$

(D) None of these

Q5 The number of solutions of the equation $|2\sin x - \sqrt{3}|^{2\cos^2 x - 3\cos x + 1} = 1$ in $[0, \pi]$ is

(A) 2 (B) 3

(C) 4 (D) 5

Q6 If $\mathrm{x} \in (0, 1)$, then greatest root of the equation $\sin 2\pi \mathrm{x} = \sqrt{2}\cos \pi \mathrm{x}$ is

(A) $\frac{1}{4}$

(B) $\frac{1}{2}$

(C) $\frac{3}{4}$

(D) $\frac{1}{3}$

Q7 The solution set of $\sin^4 x - \tan^8 x = 1$ is given by

(A) $\mathrm{x} = 2\mathrm{n}\pi + \frac{\pi}{8}, \mathrm{n} \in \mathrm{I}$

(B) $\mathrm{x} = \mathrm{n}\pi + \frac{\pi}{12}, \mathrm{n} \in \mathrm{I}$

(C) $\mathrm{x} = 2\mathrm{n}\pi + \frac{\pi}{24}, \mathrm{n} \in \mathrm{I}$

(D) None of these

Q8 Let the number of solutions of the equation $\sin x = x^2 - 4x + 5$ be λ. The general solution of the equation $1 + \tan\left(\left(\lambda^2 - 2\lambda + 1\right)\theta\right) + \frac{2}{1 - \tan\left(\left(\lambda^2 + 2\lambda + 1\right)\theta\right)} = 0$ is

(A) $\mathrm{n}\pi + \frac{\pi}{4}$ only $\mathrm{n} \in \mathrm{Z}$

(B) $\mathrm{n}\pi \pm \frac{\pi}{4}, \mathrm{n} \in \mathrm{Z}$

(C) $\mathrm{n}\pi + \frac{\pi}{3}$ only $\mathrm{n} \in \mathrm{z}$

(D) $\mathrm{n}\pi \pm \frac{\pi}{3}, \mathrm{n} \in \mathrm{Z}$

Q9 The arithmetic mean of the roots of the equation $4\cos^3 \mathrm{x} - 4\cos^2 \mathrm{x} - \cos(315\pi + \mathrm{x}) = 1$ in the interval $(0, 315)$ is equal to

(A) 50π

(B) 51π

(C) 100π

(D) 315π

Q10 The number of ordered pairs (α, β), where $\alpha, \beta \in [0, 2\pi]$ satisfying $\log_{2\sec x}\left(\beta^2 - 6\beta + 10\right) = \log_3|\cos \alpha|$ is

(A) 1 (B) 2

(C) 3 (D) 4

Q11 The number of distinct solutions of the equation $\frac{5}{4}\cos^2 2x+\cos^4 x+\sin^4 x+\cos^6 x+$ $\sin^6 x=2$ in the interval $[0,2\pi]$ is

(A) 4 (B) 8

(C) 10 (D) 12

Q12 The number of solution of the equation $1+\sin$ $x\sin^2\frac{x}{2}=0$ in $[-\pi,\pi]$, is

(A) 0 (B) 1

(C) 3 (D) none of these

Q13 One root of the equation $\cos x-x+\frac{1}{2}=0$ lies in the interval

(A) $\left[0,\frac{\pi}{2}\right]$

(B) $\left[-\frac{\pi}{2},0\right]$

(C) $\left[\frac{\pi}{2},\pi\right]$

(D) $\left[\pi,\frac{3\pi}{2}\right]$

Q14 The number of solutions of the equation $\sin x$ $=|\cos 3x|$ in $[0,\pi]$, is

(A) 3 (B) 4

(C) 5 (D) 6

Q15 If m and $n(n>m)$ are positive integers, the number of solutions of the equation $n|\sin x|=m|\cos x|$ in $[0,2\pi]$, is

(A) m

(B) n

(C) mn

(D) None of these

Answer Key

Q1	(D)
Q2	(C)
Q3	(C)
Q4	(D)
Q5	(B)
Q6	(C)
Q7	(D)
Q8	(D)
Q9	(B)
Q10	(C)
Q11	(B)
Q12	(A)
Q13	(A)
Q14	(D)
Q15	(D)